Frequency monitoring device

Background

The frequency of a power system is a critical parameter in determining the stability and reliability of the grid. It serves as an indicator of the system's balance between supply and demand. Deviations from the standard frequency can lead to significant disruptions, including large-scale blackouts. Accurate, high-resolution frequency data is essential for diagnosing and preventing these issues. However, current monitoring systems for the Great British power grid do not provide the required accuracy or real-time capabilities.

Motivation for the project

The motivation for this project is to fill the gap in real-time frequency monitoring for the Great British power grid. By developing a compact, high-precision frequency monitoring device, or a web-based data analysis tool, this project aims to provide an open data solution that can be used by researchers and industry professionals. This tool will improve the ability to diagnose issues in the power grid, potentially preventing large-scale blackouts and enhancing grid stability.

Overall aim

The primary aim of this project is to develop a system for real-time frequency monitoring of the Great British power grid. This system will either involve a hardware device that captures and transmits frequency data and a web interface that aggregates and presents this data from multiple sources.

Associated objectives (milestones) of the project:

Objective 1: Conduct research and requirements analysis to define the specifications for the frequency monitoring device.
Objective 2: Design and develop a hardware prototype for the frequency monitoring device. (software simulation + hardware implement)
Objective 3: Coding for microcontroller to calculate frequency and data storage.
Objective 4: Implement a data transmission module to upload frequency data wirelessly to an online database.
Objective 5: Test and validate the system for accuracy, real-time performance, and reliability under various conditions.
Objective 6: Optimize the design based on feedback and perform iterative improvements.